Distribution of d electrons in octahedral complexes:

The filling of electrons in the d orbitals in the presence of ligand field also follows Hund's rule. In the octahedral complexes with d^2 and d^3 configurations, the electrons occupy different degenerate t_{2g} orbitals and remains unpaired. In case of d^4 configuration, there are two possibilities. The fourth electron may either go to the higher energy e_g orbitals or it may pair with one of the t_{2g} electrons. In this scenario, the preferred configuration will be the one with lowest energy.

If the octahedral crystal field splitting energy (Δ_o) is greater than the pairing energy (P), it is necessary to cause paring of electrons in an orbital, then the fourth electron will pair up with an the electron in the t_{2g} orbital. Conversely, if the Δ_o is lesser than P, then the fourth electron will occupy one of the degenerate higher energy e_g orbitals.

For example, let us consider two different iron(III) complexes $[Fe(H_2O)_6]^{3+}$ (weak field complex; Δ_o is 14000 cm^{-1}) and $[Fe(CN)_6]^{3-}$ (Strong field complex; Δ_o is 35000 cm^{-1}). The pairing energy of Fe^{3+} is 30000 cm^{-1}. In both these complexes the Fe^{3+} has d^5 configuration. In aqua complex, the $\Delta_o < P$ hence, the fourth & fifth electrons enter e_g orbitals and the configuration is t_{2g}^3, e_g^2. In the cyanido complex $\Delta_o > P$ and hence the fourth & fifth electrons pair up with the electrons in the t_{2g} orbitals and the electronic configuration is t_{2g}^5, e_g^0.

The actual distribution of electrons can be ascertained by calculating the crystal field stabilisation energy (CFSE). The crystal field stabilisation energy is defined as the energy difference of electronic configurations in the ligand filed (E_{LF}) and the isotropic field/barycentre (E_{iso}).

$$CFSE\ (\Delta E_o) = \{E_{LF}\} - \{E_{iso}\}$$

$$= \{[n_{t_{2g}}(-0.4)+n_{e_g}(0.6)]\ \Delta_o + n_p P\} - \{n'_p\ P\}$$

Here, $n_{t_{2g}}$ is the number of electrons in t_{2g} orbitals; n_{e_g} is number of electrons in e_g orbitals; n_p is number of electron pairs in the ligand field; & n'_p is the number of electron pairs in the isotropic field (barycentre).

Calculating the CFSE for the Iron complexes

Complex: $[Fe(H_2O)_6]^{3+}$

High Spin Complex	Low spin complex
Electronic configuration in isotropic field : d5 ↑ ↑ ↑ ↑ ↑	
No. of paired electrons (n'p)= 0 ; Therefore, $E_{iso} = 0$	

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